Predicted Atmospheric and Upwelling Radiation Profile for a Stratospheric Balloon Ascent from Brussels, 2026-04-23

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ABSTRACT

A pre-flight prediction of atmospheric temperature, ozone concentration, and upwelling broadband radiation is presented for ASGARD-XV Experiment 14, a student stratospheric balloon payload carrying a Lite-On LTR390 UV/ambient-light sensor, a DFRobot SEN0321 ozone sensor, a TE MS5611 pressure/temperature sensor, and a Maxim DS3231 precision real-time clock, launching from Brussels, Belgium (50.80°N, 4.36°E) at 14:00 CEST on 23 April 2026. The ozone profile was derived from the Copernicus Atmosphere Monitoring Service (CAMS) global composition forecast (init 2026-04-20 12:00 UTC, lead +72 h), bilinearly interpolated to the launch point. Radiative transfer was computed with libRadtran 2.0.6 using the DISORT 8-stream pseudospherical solver on a 1 nm spectral grid from 250 to 1100 nm, in 1 km altitude steps from 0 to 40 km. Headline results: total ozone column **369.950 DU**, peak mixing ratio **7.61 ppmv** at **31.1 km**, solar zenith angle **38.37°**, band-integrated upwelling irradiance at 40 km of **15.6 W m⁻²** in the 280–400 nm band and **33 W m⁻²** in the 400–700 nm band.

1. Introduction

ASGARD-XV Experiment 14 is a compact autonomous data logger designed, built, and tested at The English College in Prague. It carries four sensors sampled at 1 Hz during a stratospheric balloon ascent, with records stored to a 256 kbit AT24C256 EEPROM (primary storage) and a microSD card (backup) driven by an Arduino Pro Mini running at 3.3 V / 8 MHz. The flight profile reaches a nominal float altitude of approximately 30 km.

The purpose of this note is to produce a physically grounded pre-flight prediction of the vertical structure of temperature, ozone, and ground-facing upwelling ultraviolet and visible irradiance that the payload will encounter. These reference profiles serve three functions:

- they bound the dynamic range the LTR390 UV and ALS (ambient-light) channels will see, allowing gain settings to be chosen pre-flight;
- they provide a forecast ozone column and stratospheric peak against which the SEN0321 ozone sensor data can be qualitatively compared;
- they supply a reference temperature profile (ISA 1976) against which MS5611 in-situ readings can be evaluated.

2. Methodology

2.1 Input ozone from CAMS

Ozone mass mixing ratio *q(p)* was obtained from the CAMS global atmospheric composition forecast (`cams-global-atmospheric-composition-forecasts`). The request configuration is given below:

```
init date   : 2026-04-20
init time   : 12:00 UTC
lead time   : +72 h          → valid 2026-04-23 12:00 UTC (14:00 CEST)
area        : 50.8±0.5 N, 4.36±0.5 E
levels (22) : 1000, 925, 850, 700, 600, 500, 400, 300, 250, 200,
              150, 100, 70, 50, 30, 20, 10, 7, 5, 3, 2, 1 hPa
type        : forecast
format      : netcdf
```

The raw NetCDF field has dimensions (*forecast_reference_time, forecast_period, pressure_level, latitude, longitude*). A bilinear interpolation in (latitude, longitude) to the exact launch point was performed using `xarray`'s `interp` method. Because pressure-level forecasts are produced on a 3 h time grid, the closest grid point to the 13:00 CEST launch target is 14:00 CEST, a 1 h forward offset considered negligible for stratospheric ozone.

2.2 ISA 1976 pressure–altitude mapping

For the reference temperature profile and for altitude labelling, the U.S. Standard Atmosphere 1976 was adopted, with piecewise linear temperature lapse rates:

Layer	Base altitude [km]	Lapse rate L [K km⁻¹]
Troposphere	0 – 11	−6.50
Lower stratosphere (isothermal)	11 – 20	0.00
Middle stratosphere	20 – 32	+1.00
Upper stratosphere	32 – 47	+2.80

Pressure as a function of altitude z follows from the hydrostatic balance layer by layer. For a non-isothermal layer with base (z_b, T_b, p_b):

$$p(z) = p_b \, [T_b \,/\, (T_b + L\,(z - z_b))]^{\,g\,/\,(R\,L)}$$

and for an isothermal layer:

$$p(z) = p_b \, exp[-g\,(z - z_b) \,/\, (R\,T_b)]$$

with g = 9.80665 m s⁻² and R = 287.053 J kg⁻¹ K⁻¹ (specific gas constant for dry air). The inverse $p \rightarrow z$ is derived analytically within each layer and is used to place the CAMS pressure levels on a geometric altitude axis.

2.3 Total ozone column

The ozone column density in Dobson Units (DU) is computed as

$$N_{O_3} = \int q(z) \cdot (M_{air} \,/\, M_{O_3}) \cdot n_{air}(z)\, dz$$

where $n_{air}(z) = p(z) / (k_B\, T(z))$, with $T(z)$ taken from ISA 1976, M_{air} = 28.9647 g mol⁻¹, M_{O_3} = 47.9982 g mol⁻¹, k_B = 1.380649 × 10⁻²³ J K⁻¹, and 1 DU ≡ 2.687 × 10²⁰ molec m⁻². The integral is evaluated by the composite trapezoidal rule over the CAMS pressure levels, with a rectangular pad added from the lowest level down to the surface to avoid truncating the boundary layer.

2.4 Radiative transfer

Spectral irradiance was computed with libRadtran 2.0.6 [1]. The `uvspec` input template is:

```
latitude N 50.798056
longitude E 4.357500
time 2026 04 23 12 00 00            # UTC = 14:00 CEST
atmosphere_file midlatitude_summer
mol_file O3 o3_mmr.dat mmr          # CAMS overlay
rte_solver disort
number_of_streams 8
pseudospherical
mol_abs_param crs
source solar kurudz_0.1nm.dat       # Kurucz 0.1 nm [2], mW m⁻² nm⁻¹
wavelength 250 1100
spline 250 1100 1                   # 1 nm output grid
albedo 0.05                         # land / vegetation baseline
zout_sea <SUBSTITUTED>
output_user lambda sza zout eup edir edn eglo
quiet
```

A bash driver iterates z_{out} from 0 to 40 km in 1 km steps (41 runs). For each run the solver writes a 4-column spectral file: upwelling irradiance E_{up}, direct downwelling E_{dir}, diffuse downwelling E_{dn}, and global downwelling $E_{glo} = E_{dir} + E_{dn}$. Pseudospherical mode corrects the plane-parallel direct-beam geometry for Earth curvature, which is important at high SZA but has a small (<1%) effect at our launch geometry. The DISORT 8-stream solver resolves the multiple-scattering contribution sufficiently for integrated radiometric quantities.

2.5 Band integration

At each altitude, the upwelling irradiance $E_{up}(\lambda, z)$ in mW m⁻² nm⁻¹ is numerically integrated over spectral bands:

$$E_{band}(z) = 10^{-3} \cdot \int_{\lambda_1}^{\lambda_2} E_{up}(\lambda, z)\, d\lambda \ [W\ m^{-2}]$$

where the 10⁻³ factor converts mW to W. Band definitions used throughout:

Band	Range [nm]
UV-C	250 – 280
UV-B	280 – 315
UV-A	315 – 400
UV (total)	280 – 400
Visible	400 – 700

Band	Range [nm]
Near-infrared (NIR)	700 – 1100

3. Results

3.1 Ozone profile and column

The interpolated CAMS ozone volume mixing ratio rises from order ~50 ppbv in the lower troposphere to a stratospheric peak of **7.61 ppmv** at approximately **31.1 km** (**10 hPa**), decreasing above. The vertically integrated column is **369.950 DU**, typical of mid-latitude spring. Table 1 gives the interpolated profile at all 22 CAMS pressure levels.

Table 1. CAMS ozone forecast interpolated to (50.80°N, 4.36°E), valid 2026-04-23 12:00 UTC. Altitude column is ISA 1976 pressure → height.

Pressure [hPa]	Altitude [km]	O_3 mmr [kg kg^{-1}]	O_3 [ppmv]
1000.0	0.11	8.082e-08	0.049
925.0	0.76	9.815e-08	0.059
850.0	1.46	1.068e-07	0.064
700.0	3.01	1.072e-07	0.065
600.0	4.21	1.156e-07	0.070
500.0	5.57	1.180e-07	0.071
400.0	7.19	9.776e-08	0.059
300.0	9.16	1.211e-07	0.073
250.0	10.36	1.525e-07	0.092
200.0	11.78	3.135e-07	0.189
150.0	13.61	1.218e-06	0.735
100.0	16.18	1.760e-06	1.062
70.0	18.44	3.475e-06	2.095
50.0	20.58	5.601e-06	3.377
30.0	23.85	7.331e-06	4.421
20.0	26.48	8.899e-06	5.366
10.0	31.05	1.262e-05	7.610
7.0	33.45	1.220e-05	7.359
5.0	35.78	1.118e-05	6.744
3.0	39.43	9.132e-06	5.506
2.0	42.44	7.137e-06	4.304
1.0	47.82	4.133e-06	2.492

3.2 Upwelling radiation profile

The solar zenith angle at the launch coordinates and time is **38.37°** (elevation 51.6°). Band-integrated upwelling irradiance at nine representative altitudes is listed in Table 2.

Table 2. Band-integrated upwelling irradiance E_{up} (W m^{-2}) at representative altitudes. Pressure from ISA 1976.

z [km]	p [hPa]	UV-A 315–400	UV-B 280–315	UV 280–400	Visible 400–700	NIR 700–1100
0	1013.25	2.07	0.000	2.07	14.24	13.96
5	540.20	8.57	0.000	8.57	23.73	15.06
10	264.36	12.08	0.003	12.08	28.21	15.61
15	120.45	13.99	0.023	14.01	30.75	15.92
20	54.75	14.86	0.061	14.92	32.00	16.07
25	25.11	15.22	0.099	15.32	32.57	16.14
30	11.72	15.37	0.128	15.50	32.81	16.18
35	5.59	15.43	0.146	15.58	32.92	16.19
40	2.78	15.46	0.157	15.62	32.98	16.20

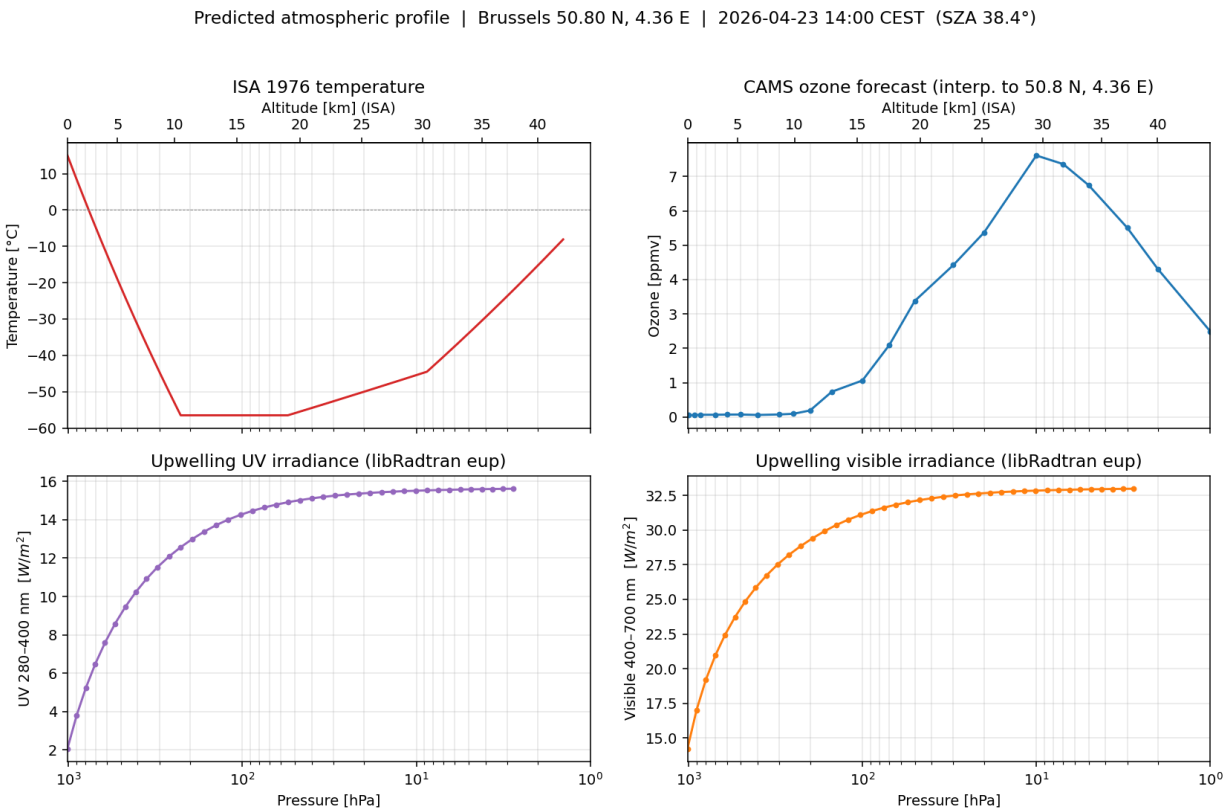

Figure 1. Predicted atmospheric and upwelling radiation profile for the ASGARD-XV Experiment 14 flight. Top-left: ISA 1976 temperature (°C). Top-right: CAMS ozone volume mixing ratio (ppmv). Bottom-left: band-integrated upwelling UV irradiance 280–400 nm (W m^{-2}). Bottom-right: band-integrated upwelling visible irradiance 400–700 nm (W m^{-2}). Shared pressure axis (log, surface on left, TOA on right). Top row shows a secondary altitude axis.

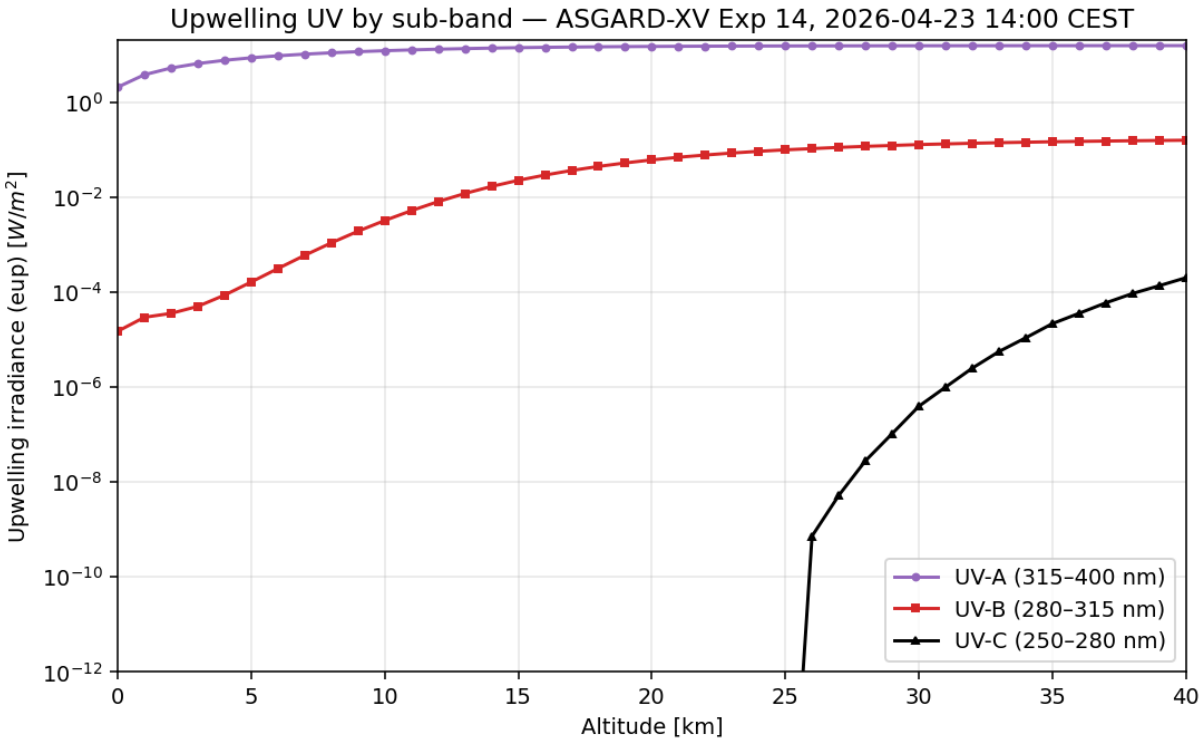

Figure 2. Upwelling UV irradiance decomposed into UV-A (315–400 nm), UV-B (280–315 nm), and UV-C (250–280 nm) components, log scale. UV-A dominates at all altitudes; UV-B is reduced by ozone absorption below the stratospheric layer; UV-C is effectively extinguished below ~25 km.

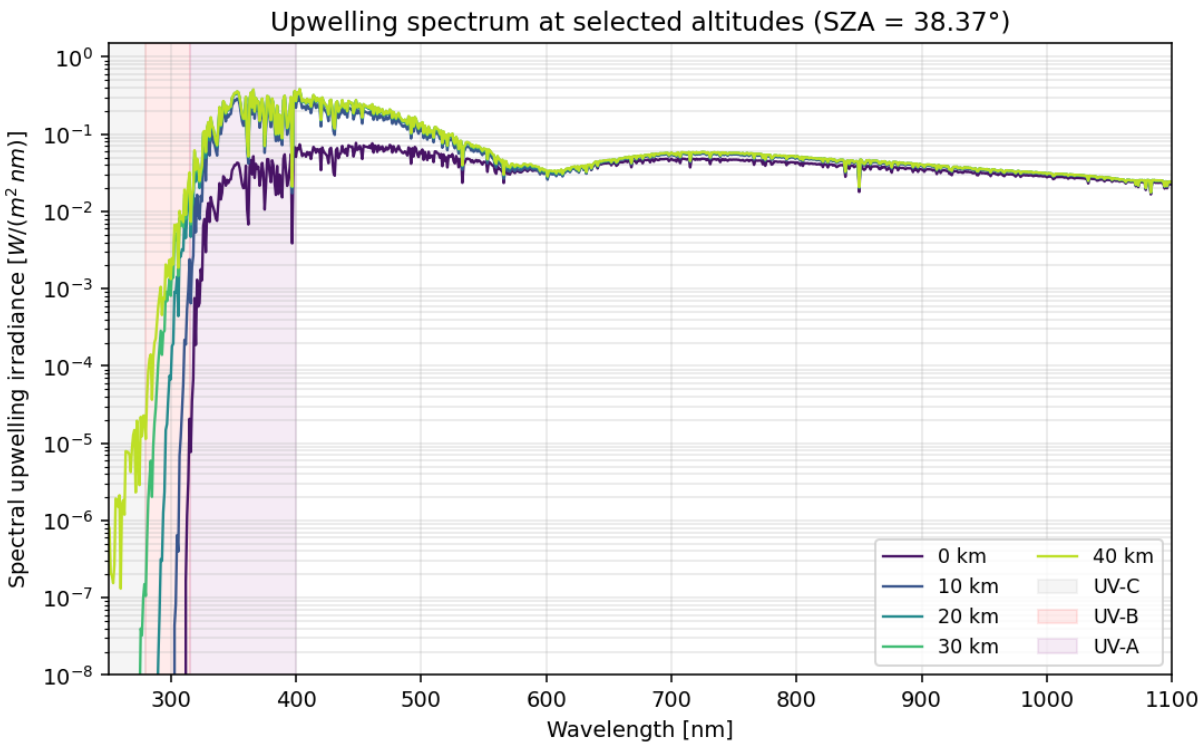

Figure 3. Spectral upwelling irradiance $E_{up}(\lambda)$ at five altitudes, log scale. The short-wavelength cut-off imposed by the Hartley-band ozone absorption migrates from ~305 nm at the surface to ~260 nm at 40 km as the overlying ozone column decreases. Shaded regions mark the UV-C / UV-B / UV-A sub-bands.

3.3 Downwelling reference

Although the primary interest of the ground-facing instrument configuration is the upwelling flux, the same libRadtran runs provide the downwelling global irradiance E_{glo} for context. Table 3 lists E_{glo} in the same UV and visible bands.

Table 3. Band-integrated downwelling (global) irradiance E_{glo} (W m^{-2}) at representative altitudes; full-spectrum 250–1100 nm column for reference.

z [km]	UV 280–400	Visible 400–700	250–1100 total
0	41.32	284.84	605.31
5	49.68	325.50	658.28
10	56.56	368.50	711.66
15	62.44	394.84	745.80
20	67.17	406.26	762.75
25	70.94	411.01	771.59
30	74.04	413.11	776.93
35	76.55	414.07	780.48
40	78.39	414.49	783.07

4. Discussion

4.1 Sanity checks

At 40 km altitude, the total downwelling irradiance in the 250–1100 nm range is **783.074 W m^{-2}**. The TOA horizontal solar flux at SZA = 38.37° would be **TSI · cos(SZA) = 1067.094 W m^{-2}**; the 250–1100 nm window retains roughly 80% of the total solar constant, consistent with the Kurucz reference spectrum. The 40 km figure thus accounts for ~73.4% of the full-spectrum horizontal TOA flux, physically reasonable given (i) the 250–1100 nm window, (ii) Rayleigh absorption and scattering in the thin overlying column, and (iii) residual stratospheric ozone absorption of the UV-B/UV-C contribution.

The upwelling-to-downwelling ratio confirms the expected wavelength dependence of Rayleigh backscattering ($\sigma \propto \lambda^{-4}$):

Altitude	UV (280-400) E_{up}/E_{glo}	Visible (400-700) E_{up}/E_{glo}
0 km (surface)	5%	5%
40 km	19.9%	7.96%

At the surface both ratios are dominated by the imposed surface albedo of 0.05 and thus are similar. At 40 km the UV ratio is substantially larger than the visible because the entire atmospheric column contributes diffuse Rayleigh backscatter, which is strongly wavelength-dependent.

4.2 Limitations

- **Cloud-free assumption.** The simulation does not include clouds. Cloud cover in the trajectory corridor would (a) substantially increase upwelling visible irradiance above the cloud deck (bright reflector), and (b) reduce UV below the deck. The predicted curves apply to a cloud-free case and should be treated as envelopes where cloud is present.
- **Surface albedo.** A constant albedo of 0.05 (generic land/vegetation) is used. Trajectory-dependent mosaics of land, water, snow, or cloud-top would require a rescaling; for the upwelling band, $E_{up}(z)$ scales roughly linearly with the albedo of the underlying surface within the cone subtending the solid angle at altitude.
- **Aerosol.** libRadtran's default rural aerosol model is applied. Actual boundary-layer aerosol optical depth over NW Europe in late April can vary; the effect on UV-B and visible irradiance is at the several-percent level and does not change qualitative conclusions.
- **CAMS time grid.** Pressure-level CAMS output is on a 3 h time grid; the closest point to the 13:00 CEST target is 14:00 CEST (+1 h). Stratospheric ozone varies on much longer timescales, so this offset is negligible.
- **Temperature inconsistency.** Figure 1(a) shows ISA 1976 temperature for vertical coordinate consistency, while the radiative transfer internally uses libRadtran's `midlatitude_summer` profile (temperature differences <3 K in the troposphere, sub-percent impact on integrated UV).
- **SEN0321 at stratospheric density.** The DFRobot SEN0321 electrochemical ozone sensor is calibrated for surface-level pressures; at stratospheric densities (< 30 hPa) its response is outside the manufacturer-specified operating regime and the sensor data should be treated as qualitative above ~20 km.

4.3 Implications for the in-flight measurement

The LTR390 UV channel has a peak responsivity near 330 nm, falling within the UV-A sub-band. Ground-facing during ascent, it should see an upwelling UV-A irradiance rising from **2.07 W m⁻²** at launch to **15.5 W m⁻²** at 40 km (Table 2). The ALS (visible) channel, peaking near 555 nm, should see upwelling visible irradiance from **14.2 W m⁻²** to **33 W m⁻²**. Both channels will therefore span roughly one order of magnitude over the flight, and gain settings should be chosen to avoid saturation near the float altitude while retaining sensitivity at launch.

5. Conclusion

A complete pre-flight prediction of the atmospheric and upwelling radiative environment for ASGARD-XV Experiment 14 has been produced, combining a 72-hour CAMS ozone forecast with libRadtran DISORT radiative transfer. The headline figures are a total ozone column of **369.950 DU**, a stratospheric ozone peak of **7.61 ppmv at 31.1 km**, and upwelling irradiance at 40 km of **15.6 W m⁻²** (UV 280–400 nm) and **33 W m⁻²** (visible 400–700 nm). All input files, scripts, and outputs are archived in the experiment repository and can be re-executed with the `cams_download3.py` → `interpolation1.py` → `run_loop.sh` → `paper.py` pipeline. The predictions provide a quantitative reference against which the LTR390, SEN0321, and MS5611 sensor data will be compared after flight.

References

1. Emde, C., Buras-Schnell, R., Kylling, A., Mayer, B., Gasteiger, J., Hamann, U., Kylling, J., Richter, B., Pause, C., Dowling, T., Bugliaro, L. (2016). The libRadtran software package for radiative transfer calculations (version 2.0.2). *Geoscientific Model Development*, **9**, 1647–1672. doi:10.5194/gmd-9-1647-2016.
2. Kurucz, R. L. (1992). Synthetic infrared spectra. In *Infrared Solar Physics*, IAU Symposium 154, ed. D. M. Rabin et al., Kluwer Academic Publishers, 523–531.
3. U.S. Committee on Extension to the Standard Atmosphere (1976). *U.S. Standard Atmosphere, 1976*. NOAA-S/T 76-1562. U.S. Government Printing Office, Washington, DC.
4. Copernicus Atmosphere Monitoring Service (CAMS), Atmosphere Data Store. *Global atmospheric composition forecasts*. European Centre for Medium-Range Weather Forecasts. `https://ads.atmosphere.copernicus.eu` (accessed April 2026).
5. Stamnes, K., Tsay, S.-C., Wiscombe, W., Jayaweera, K. (1988). Numerically stable algorithm for discrete-ordinate-method radiative transfer in multiple scattering and emitting layered media. *Applied Optics*, **27**, 2502–2509.